

WebSocket FT4/8 v1.0

WebSocket FT4/8 is a self-contained FT8 and FT4 decoder that runs entirely in a browser with no installation required. It connects to SDRplay's SDRconnect software via WebSocket and decodes FT8 and FT4 signals in real time. Select a band, adjust the audio level, and click Decode to begin. Decoded messages are displayed after each cycle and can be explored in a split window. A live spectrum and waterfall display the signal environment, and a station map plots decoded callsigns on a world map for the session.

Features

- **FT8 and FT4 decoding**
 - Pure browser-based decoder — no external software or plugins required.
 - **FT8 / FT4** — switch between modes at any time. The current mode is shown on the button.
 - **Fast / Normal / Deep** — sets the decoder depth. Fast uses minimal CPU; Normal is the recommended default; Deep attempts more aggressive decoding at the cost of higher CPU usage.
 - **Country display** — optionally show the country alongside decoded callsigns. Three modes: Off, CQ (CQ messages only), or All (every decode).
- **Decode window**
 - Messages are displayed after each decode cycle showing UTC time, dB, DT, frequency, and message text.
 - **Left-click any message** — opens a split window showing all messages decoded on that frequency.
 - **Right-click any message** — opens the sender's QRZ page in a new tab. This works in both the main decode window and the split window.
 - **Clear** — clears the decoded messages list and map display.
- **Band selection**
 - Eleven amateur radio bands supported: 160m, 80m, 60m, 40m, 30m, 20m, 17m, 15m, 12m, 10m, and 6m.
 - Selecting a band resets the decoder and retunes SDRconnect to the correct frequency.
- **Band hopping**
 - Automatically cycles through up to 11 bands in sequence.
 - Each band can be configured individually with its own cycle time and FT8/FT4

mode.

- When hopping is active, the Decode, Mode, and Band controls are locked.
- The hopping modal shows current band, next hop, cycle position, and session decode totals.
- Each band tracks its own decode, callsign, and country counts, with a Reset Stats button to clear them.
- Your band hopping configuration is saved automatically and restored the next time you open the app.

- **Station map**

- All decoded stations for the session are plotted on a world map.
- Stations are located using their transmitted Maidenhead grid square when available, otherwise by country.
- Map markers are colour-coded by band. Click any marker to see station details, including a QRZ button beside the callsign for resolved stations that opens the station's QRZ page in a new tab.
- Zoom in and out using the mouse wheel. Pan by holding left-click and dragging.
- The map resets when the decode panel is cleared, when disconnecting, or when changing bands.

- **Spectrum and waterfall**

- Live spectrum display shows signal levels across the FT8/FT4 audio passband (0–3320 Hz).
- Waterfall shows signal history over time, with stronger signals appearing brighter.
- **Slow / Fast** — sets the waterfall scroll speed.
- **Cumulative** — toggles frame averaging for a smoother spectrum display.
- **Autoscale** — automatically adjusts the Min and Max dB levels to fit the current signal environment.
- **Min / Max** — set the dB range of the colour scale manually using the input fields or ▲ ▼ buttons.
- **Clear** — clears the spectrum and waterfall display.

- **Audio level and metering**

- Volume slider and mute button control the audio output level.
- Audio Meter shows the input signal level. Adjust the Audio Level so the meter

shows green bars to mid-scale with no orange or red bars.

- **Filter bandwidth**

- Sets the SDRconnect receiver filter bandwidth in Hz.
- Adjustable by typing a value directly or using the ▲ ▼ step buttons (100 Hz steps, 0–3300 Hz).

- **AGC**

- Enables or disables the Automatic Gain Control in SDRconnect.
- The recommended setting is disabled (AGC Off).

- **Signal displays**

- **S-Meter** — graphical signal strength display from S1 to S9+60 dB.
- **Signal Power** — numeric signal power reading from SDRconnect.
- **SNR** — signal-to-noise ratio.
- **Frequency** — current VFO frequency as reported by SDRconnect.

- **WebSocket connection**

- Connects to SDRplay's SDRconnect via WebSocket.
- Default connection: 127.0.0.1 port 5454 (both configurable).
- IP address field remembers previously entered addresses and shows them as a drop-down history.

Requirements

- **SDRconnect** — SDRplay's SDRconnect software must be running on the same machine or local network with WebSocket access enabled.
- **Supported browsers** — Google Chrome, Microsoft Edge, or Firefox.
- **Accurate system clock** — for correct FT8/FT4 decoding your system clock must be synchronized to UTC within 1 second. Windows users should run a dedicated NTP sync tool such as Nettime or Dimension 4.
- **Internet connection required for the station map** — the map uses MapLibre GL and OpenFreeMap tiles, both loaded from the internet. All other features work without an internet connection.
- No installation and no server required.

Typical Workflow

1. **Open WebSocket FT4/8** in your browser. Just double-click the file.

2. Enter the SDRconnect server IP address (default **127.0.0.1**) and port (default **5454**), then click **Connect**.
3. Adjust the **Audio Level** slider so the Audio Meter shows green bars to mid-scale with no orange or red bars.
4. Select a **Band** from the drop-down to tune SDRconnect to the correct frequency.
5. Click **Decode Off** to start decoding. The button turns green when active.
6. Decoded messages appear in the decode window after each cycle. Left-click any message to open the split window for that frequency.
7. Use **Country Off / CQ / All** to optionally display the country alongside decoded callsigns.
8. Click **Map** to open the station map and see decoded callsigns plotted on a world map.
9. To scan multiple bands automatically, click **Hop Off** to open the Band Hopping modal, configure your bands and cycle times, then enable hopping.
10. Adjust **Min** and **Max** or click **Autoscale** to optimise the spectrum and waterfall display for your signal environment.

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